Network Traffic Analyzer

A Python project for summarizing network traffic

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Why I built it

Packet lists are overwhelming and hard to make sense of.

My app summarizes traffic so patterns are easy to see.

It highlights rule-based activity for human review.

How the app works

Capture

Scapy captures visible packet metadata

Process

```
import pandas as pd

df = pd DataFrame(data)

summary = (df .groupby('Source IP',
'Protocol') .
 .agg(
   _Packets=('Length', 'count') ,
   _Bytes=('Length', '_sum')
   )
 .resetindex())
```

Pandas groups and summarises rows

Store

MySQL saves capture sessions and packet rows

Visualize

Streamlit and Plotly display charts and exports

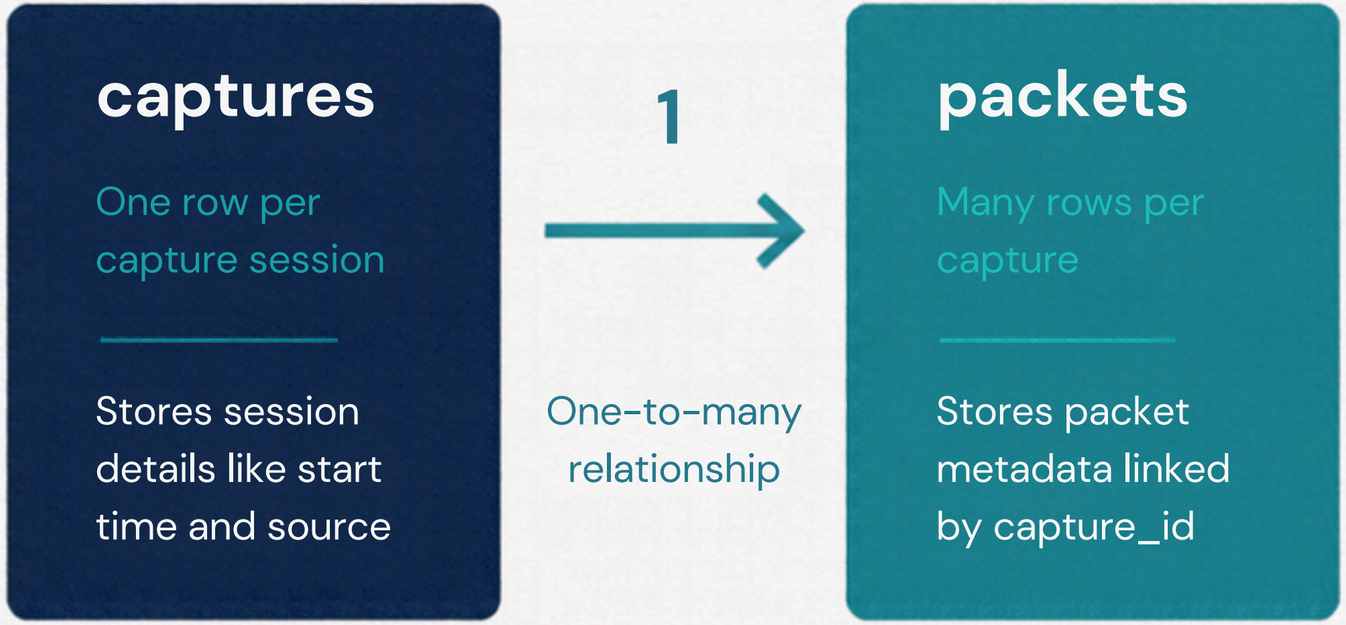
Tools and data model

Tech stack

Python, Scapy, Pandas, Streamlit, Plotly, MySQL Connector/Python

Data model

Built on a simple, relational structure.



captures records each capture session.

packets stores many packet metadata records linked to a capture via capture_id.

One capture has **many** packets.

What the dashboard shows

01 At-a-glance totals

See packet, byte, source-device, and protocol totals at a glance.

02 Visual insights

Explore protocol mix, packet timeline, traffic direction, and top sources.

03 Detailed packet table

Inspect each packet with timestamp, IPs, ports, protocol, and length.

How alerts work

Rules flag high packet counts
or byte volume

Other rules check many destination
ports, packet rate, repeated TCP SYNs,
and unrecognized local sources

An alert is a lead for review,
not proof of an attack

Results and reports

20
PACKETS

Tested a live capture of 20 IP packets on an authorized device.

476
ROWS

Synthetic demonstration contained 476 rows: 364 outgoing and 112 incoming.

MySQL save/reload worked; the app exports a traffic summary and packet CSV.

Limits and what I learned

LIMITS

Capture sees only traffic available to the interface.

Payloads aren't stored and alerts may be false positives.

Use live capture only on authorized devices and networks.

WHAT I LEARNED

How packet metadata reveals patterns.

Building traffic summaries from live data.

Persisting data for reliable analysis.

Turning insights into clear dashboard reports.